

IROHA: Three-Phase Gradual Pruning for 3D Gaussian Splatting Compression

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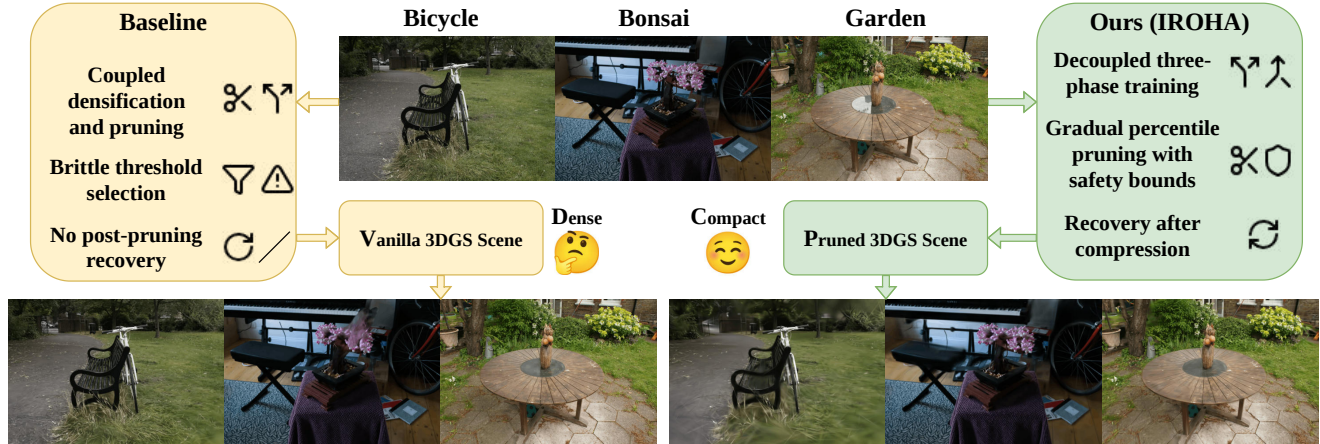


Figure 1: Visual overview of scene compression in IROHA. The framework first learns geometry, then prunes progressively, and finally recovers image quality through fine-tuning.

ABSTRACT

3D Gaussian Splatting (3DGS) delivers real-time, photorealistic novel-view synthesis but routinely produces scenes with hundreds of thousands to millions of Gaussian primitives, creating storage and deployment bottlenecks. Existing pruning methods typically rely on hand-tuned thresholds or one-shot deletion heuristics, making the quality-compression trade-off brittle and hard to recover once important Gaussians are removed.

We propose IROHA, a three-phase gradual pruning architecture that decouples geometry learning, controlled active-set compression, and recovery into distinct stages. Phase 1 learns a high-quality scene with standard 3DGS densification. Phase 2 freezes densification and performs gradual percentile pruning: at regular intervals, a bounded number of low-score Gaussians is removed under per-event and global keep-floor constraints, while the surviving set re-optimises between deletion events. Phase 3 fine-tunes the compressed model with pure photometric loss.

Experiments on NeRF Synthetic show that IROHA removes 35–45% of Gaussians while preserving or improving quality under a matched training budget. Against vanilla 3DGS trained for the same 40K iterations, IROHA improves PSNR/SSIM/LPIPS from 31.30/0.958/0.056 to 31.61/0.961/0.053 while using 40% fewer Gaussians. A post-training 8-bit quantization stage retains 31.45/0.959/0.055 at 5.5× total compression. On Mip-NeRF 360, eight paired scenes maintain visual quality on average (mean +0.05 dB)

at ~42% fewer Gaussians, while the separate *kitchen* report uses the summary-last protocol. These results suggest that a substantial portion of 3DGS redundancy is structured, and that a recovery-aware pruning framework provides an effective mechanism to exploit this structure, even with a simple opacity-based scoring proxy.

CCS CONCEPTS

• Computing methodologies → Computer graphics; Neural networks.

KEYWORDS

3D Gaussian Splatting, Model Compression, Pruning, Gradual Pruning, Novel View Synthesis

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1 INTRODUCTION

3D Gaussian Splatting (3DGS) [25] represents scenes as collections of learnable anisotropic Gaussians that are rasterised in a single forward pass, achieving real-time rendering at quality competitive with Neural Radiance Fields [32]. However, the adaptive densification strategy routinely produces 10^5 – 10^7 primitives, resulting in large model sizes and heavy deployment costs [14, 28].

Existing pruning methods remove Gaussians whose opacity, gradient magnitude, or rendering contribution falls below a hand-tuned threshold [14, 15, 19]. These approaches share

two limitations: (i) the threshold must be chosen carefully—too low and almost nothing is pruned, too high and quality collapses—and (ii) pruning is often treated as an instantaneous decision rather than as a process with recovery. In 3DGS, this is particularly problematic because the model is highly over-parameterised locally: many nearby Gaussians cooperate to explain high-frequency appearance, so deleting them too abruptly can damage thin structures and occlusion boundaries before the optimiser has time to redistribute responsibility.

We identify a further structural problem: in standard 3DGS training, densification and pruning operate simultaneously, creating a conflict where new Gaussians are added in the same regions where others are being removed. This observation motivates a *decoupled* approach in which geometry growth, active-set reduction, and post-pruning recovery are handled in separate phases.

Our key contribution is a principled architectural formulation that recasts pruning as a recovery-aware schedule design problem. Once Phase 2 starts, the live set shrinks monotonically, each prune step is budgeted, and recovery time becomes an explicit part of the compression design. This architectural viewpoint explains why gradual percentile pruning stays stable under aggressive compression and why schedule-recovery interaction unlocks strong gains even with simple scoring. This suggests that pruning stability in 3DGS is governed more by recovery dynamics than by the specific importance metric.

Contributions.

- A **three-phase pruning architecture** for 3DGS compression that separates geometry learning, gradual pruning, and post-pruning recovery, avoiding the densification-pruning conflict.
- A **budgeted pruning controller** based on opacity-ranked percentile deletion with per-event caps and a keep floor, turning threshold-sensitive pruning into a directly controllable schedule.
- **Empirical evidence** that, in our unified same-codebase setting, recovery-aware scheduling plays a dominant role in compression performance, enabling strong results with standard opacity-based scoring while revealing structured, local 3DGS redundancy.

Compressed 3DGS assets matter for web-based viewers, mixed-reality clients with tight memory budgets, and collaborative tools where large checkpoints must be synchronised frequently. Since each Gaussian stores 59 float32 parameters (~236 bytes), reducing primitive count by 40% directly translates to proportional savings in storage, bandwidth, and rendering memory—before any additional quantisation or coding is applied.

2 RELATED WORK

2.1 Novel View Synthesis

NeRF [32] popularised coordinate-based neural scene representations optimised via differentiable volume rendering. Subsequent work improved quality [3, 4], training speed [7, 18, 35, 42], and generalisation [50]. 3DGS [25] departs from the implicit paradigm by representing scenes with explicit anisotropic Gaussians rasterised via tile-based splatting, achieving real-time rendering with competitive quality. Several extensions improve geometry, aliasing, or scene structure, including 2DGS [22], Scaffold-GS [31], Mip-Splatting [51], Octree-GS [40], and GaussianPro [12].

2.2 3DGS Compression and Pruning

Several concurrent works tackle the storage overhead of 3DGS. Pruning-oriented methods remove Gaussians judged unimportant: LightGaussian [14] uses a global significance score combining opacity and volume; Mini-Splatting [15] applies importance-based simplification; EAGLES [19] scores sensitivity to quantisation; Reduced-3DGS [38] ranks primitives by rendering contribution; and PUP 3D-GS [20] studies uncertainty-aware pruning. Parameter compression methods focus on how to encode retained Gaussians more efficiently, including Compact-3DGS [28, 36], HAC/HAC++ [10, 11], and Compressed 3DGS [37]. More recently, LP-3DGS [53] learns a pruning mask and optimal prune ratios; SA-3DGS [52] learns importance scores for adaptive pruning; Clean-GS [33] leverages semantic masks to guide pruning; FlexGaussian [44] proposes training-free compression; and a comprehensive survey [2] frames the growing design space. Additional recent designs span uncertainty-driven pruning [1], full-factor importance scoring [49], hierarchical/context-aware compression [13, 23], spherical-Gaussian pruning [8], descriptor-based pruning [16], feed-forward compression [9], confidence-based pruning [39], and adaptive pruning with Difference-of-Gaussian primitives [46]. Our method is narrower: it keeps the opacity proxy fixed and isolates how much compression can be recovered from schedule design alone.

2.3 Variational and Probabilistic Regularisation

Variational inference [5, 24] approximates intractable posteriors by minimising a KL-regularised objective, while the reparameterisation trick of VAEs [26, 41] enables scalable gradient-based optimisation. Sparse variational methods and Bayesian compression schemes have long used KL penalties or sparsity-inducing priors to suppress redundant parameters [6, 29, 30, 34]. In the Gaussian Splatting literature, recent probabilistic or regularisation-based formulations include VBGS [45], SVR-GS [43], StableGS [47], TSPE-GS [48], and Confident Splatting [39]. Our focus is narrower: we variationalise only opacity inside a fixed three-phase schedule rather than learning a separate confidence model for pruning. Prior work largely emphasizes scoring or

coding; our formulation instead treats schedule as a first-class variable in a constrained, recovery-aware optimization process.

2.4 Gradual Pruning as a Schedule Design

Iterative magnitude pruning (IMP) [17] shows that sparsity is often easier to obtain through repeated small deletions with retraining in between than through one-shot thresholding. We borrow this schedule intuition for 3DGS primitives: the key role of graduality here is not only better final sparsity, but also the chance for the renderer to partially recover between pruning events.

3 METHODOLOGY

Figure 2 summarises the high-level pipeline that combines geometry learning, gradual percentile pruning, and post-pruning recovery. The key distinction from the original one-stage formulation is that compression is isolated from geometry growth and followed by a dedicated recovery phase.

3.1 Preliminaries

A 3DGS scene consists of N Gaussians, each with mean $\mu_i \in \mathbb{R}^3$, covariance Σ_i , opacity $\alpha_i = \text{sigmoid}(o_i) \in (0, 1)$, and SH colour coefficients $\mathbf{c}_i$. Rendering α -composites projected Gaussians front-to-back:

$$\hat{C}(\mathbf{x}) = \sum_{i \in \mathcal{N}(\mathbf{x})} \mathbf{c}_i \omega_i(\mathbf{x}) \prod_{j < i} (1 - \omega_j(\mathbf{x})), \quad (1)$$

with $\omega_i = \alpha_i \cdot G_i(\mathbf{x})$ the 2D Gaussian weight and $j < i$ denoting front-to-back depth order. Training minimises $\mathcal{L} = (1-\lambda)\mathcal{L}_1 + \lambda\mathcal{L}_{\text{D-SSIM}}$ with $\lambda=0.2$.

To reason about compression explicitly, we introduce a binary activity mask $\mathbf{m} \in \{0, 1\}^N$, where $m_i = 1$ indicates that Gaussian i is active. The masked renderer becomes

$$\hat{C}(\mathbf{x}; \Theta, \mathbf{m}) = \sum_{i \in \mathcal{N}(\mathbf{x})} m_i \mathbf{c}_i \omega_i(\mathbf{x}) \prod_{j < i} (1 - m_j \omega_j(\mathbf{x})). \quad (2)$$

This view makes pruning a discrete active-set decision rather than a purely heuristic parameter edit: only Gaussians with $m_i = 1$ contribute to the accumulated colour and transmittance chain, while masked-out Gaussians are skipped by the compositor.

3.2 Recovery-Aware Compression Objective

Let Θ denote all Gaussian parameters and let $\mathbf{m}^{(k)} \in \{0, 1\}^{N_0}$ be the active-set mask after the k -th prune event in Phase 2, with $N_k = \|\mathbf{m}^{(k)}\|_0$ and N_0 the number of active Gaussians at the start of compression. The ideal compression problem after Phase 1 can be written as

$$\begin{aligned} \min_{\Theta, \mathbf{m}} \quad & \mathcal{L}_{\text{photo}}(\Theta, \mathbf{m}) + \lambda_c \frac{\|\mathbf{m}\|_0}{N_0} \\ \text{s.t.} \quad & \mathbf{m} \preceq \mathbf{m}^{(k)}, \\ & \|\mathbf{m}^{(k)} - \mathbf{m}\|_0 \leq B_k, \\ & \|\mathbf{m}\|_0 \geq R_{\min}, \end{aligned} \quad (3)$$

where $\mathbf{m} \preceq \mathbf{m}^{(k)}$ enforces *monotone pruning* (once removed, a Gaussian cannot re-enter during Phase 2), B_k is the per-event deletion budget, and $R_{\min}$ is the global keep floor. The first term preserves image quality, while the second encourages a compact active set. This highlights pruning in 3DGS as a coupled discrete-continuous optimization problem, where discrete update timing directly affects recoverability.

Exact optimisation of (3) is combinatorial. Our practical solver therefore alternates between (i) a *discrete projection step* that updates the active set under the current budget and (ii) continuous gradient-based optimisation that lets the surviving Gaussians absorb the induced reconstruction error. This alternating view explains why the recovery horizon is a first-class part of the method rather than a cosmetic fine-tuning stage.

3.3 Three-Phase Schedule and Greedy Projection

The key insight is that densification (which *adds* Gaussians) and compression pruning (which *removes* them) should not operate simultaneously. We therefore split training into three phases.

Phase 1: Geometry learning (iterations $0 \rightarrow T_s$). Standard 3DGS training with unrestricted densification, opacity reset, and the default photometric loss. This phase produces the high-quality scene that defines the initial active set $\mathbf{m}^{(0)}$ for compression. We set $T_s = 15,000$.

Phase 2: Gradual compression (iterations $T_s \rightarrow T_c$). Densification is frozen. Every D_p iterations, we apply a budgeted pruning event. The number of Gaussians that may be removed at event k is

$$B_k = \min\left(\lfloor PN_k \rfloor, \lfloor F_{\max} N_k \rfloor, N_k - R_{\min}\right), \quad (4)$$

where P is the target prune ratio, $F_{\max}$ is the per-event safety cap, and $R_{\min}$ is the global keep floor. For all reported main results, the scalar importance score is

$$s_i^{\text{base}} = \text{sigmoid}(o_i). \quad (5)$$

This is the same quantity as opacity α_i from Section 3.1; the key design is to embed this proxy inside the three-phase pruning architecture, turning opacity into a directly controllable compression signal. The indices removed at event k are the lowest-score active Gaussians:

$$\mathcal{P}_k = \text{BottomK}_{B_k} \{s_i^{\text{base}} \mid m_i^{(k)} = 1\}. \quad (6)$$

The mask update is

$$m_i^{(k+1)} = \begin{cases} 0, & i \in \mathcal{P}_k, \\ m_i^{(k)}, & \text{otherwise.} \end{cases} \quad (7)$$

Between pruning events, all remaining Gaussians continue training with the photometric loss, allowing the model to partially compensate for removed primitives before the next round. In practice, the D_p -step interval after each prune event lets the surviving set absorb local reconstruction error

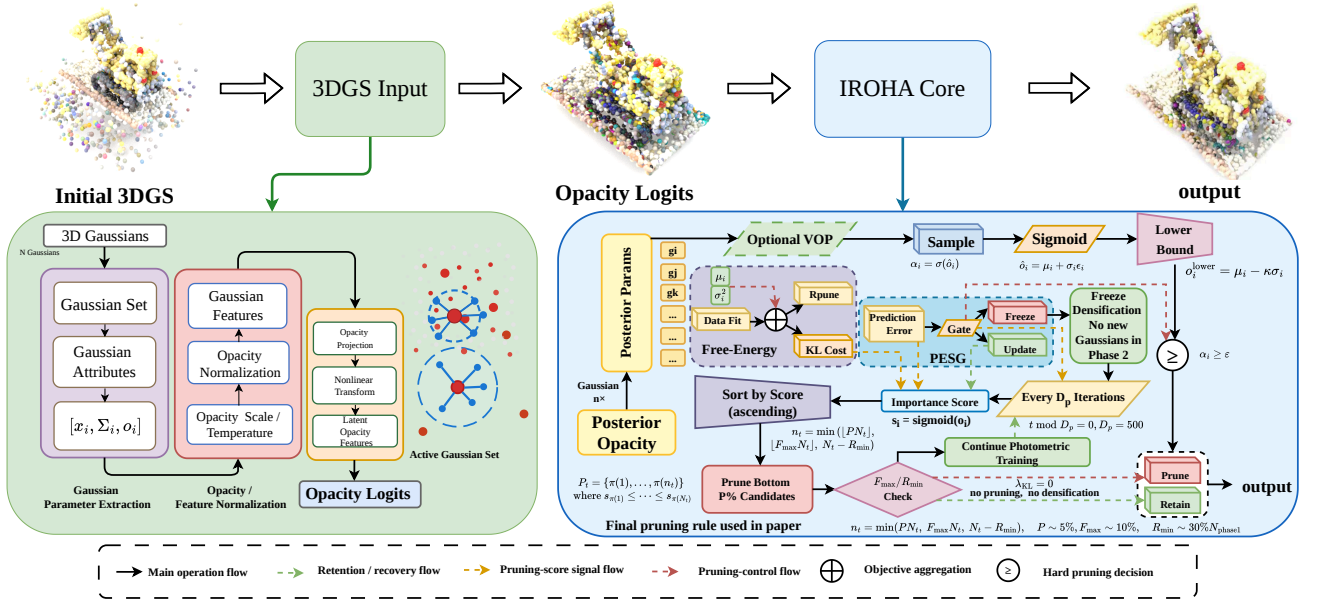


Figure 2: Overview of IROHA as a three-phase pruning architecture. Phase 1 learns geometry with standard 3DGS densification, Phase 2 freezes densification and performs gradual percentile pruning, and Phase 3 fine-tunes the pruned model to recover quality. Dashed paths indicate ancillary diagnostic branches attached to the main architecture.

before the next compression step arrives, and Phase 3 handles the remaining recovery. We set $T_c = 25,000$, $D_p = 500$, giving up to 20 pruning events.

Phase 3: Quality recovery (iterations $T_c \rightarrow T_{\max}$). No pruning, no densification. Pure photometric fine-tuning of all remaining Gaussian parameters:

$$\min_{\Theta} \mathcal{L}_{\text{photo}}(\Theta, \mathbf{m}^{(K)}), \quad (8)$$

where K is the final prune event. This phase is critical: our experiments show that sufficient recovery is the key to retaining—and even improving—quality after pruning. We set $T_{\max} = 40,000$ (15,000 recovery steps).

3.4 Brief VOP Diagnostic

We also study a probabilistic opacity branch as an auxiliary uncertainty-analysis extension. The main operating point reported in the paper uses the deterministic three-phase architecture, while this branch augments each Gaussian with a posterior over logit-opacity. Specifically, we write

$$\tilde{o}_i = \mu_i + \sigma_i \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 1), \quad \alpha_i = \text{sigmoid}(\tilde{o}_i). \quad (9)$$

The corresponding posterior is $q_i = \mathcal{N}(\mu_i, \sigma_i^2)$, while the prior is $p_i = \mathcal{N}(\mu_0, \sigma_0^2)$ with $\mu_0 = -2$ and $\sigma_0 = 0.1$. The KL penalty is

$$D_{\text{KL}}(q_i \| p_i) = \frac{1}{2} \left[\frac{\sigma_i^2 + (\mu_i - \mu_0)^2}{\sigma_0^2} - 1 - \log \frac{\sigma_i^2}{\sigma_0^2} \right], \quad (10)$$

and the Phase 2 objective becomes

$$\mathcal{F} = \mathcal{L}_{\text{photo}} + \beta \sum_i D_{\text{KL}}(q_i \| p_i). \quad (11)$$

In the current implementation, rendering uses a *deterministic mean-forward pass* with $\tilde{o}_i = \mu_i$; σ_i enters through the KL term and the uncertainty-analysis score below. With the present low-opacity prior ($\mu_0 = -2$, i.e. $\alpha \approx 0.12$), the KL term suppresses opacity logits too broadly and hurts aggressive pruning, so the strongest compression operating point remains the deterministic base-score branch in (5). VOP therefore serves as an auxiliary uncertainty-analysis branch alongside the main architecture. It also enables an uncertainty-aware *analysis score*

$$s_i^{\text{lb}} = \text{sigmoid}(\mu_i - \kappa \sigma_i), \quad (12)$$

which acts as a lower credible bound on opacity. We use this score for diagnostics and qualitative pruning analysis.

3.5 Schedule Properties

The discrete update in (7) has two immediate consequences. First, the active-set size decreases monotonically and never crosses the keep floor:

$$N_{k+1} = N_k - B_k, \quad R_{\min} \leq N_{k+1} \leq N_k. \quad (13)$$

Second, when the per-event cap and keep floor are inactive, $B_k \approx PN_k$ and the count follows the approximate geometric envelope

$$N_k \approx (1 - P)^k N_0. \quad (14)$$

This back-of-the-envelope relation is useful as a schedule-design rule of thumb: it estimates how many pruning events are needed to reach a target compression regime before recovery saturates.

Algorithm 1 IROHA Three-Phase Training

Require: Images $\mathcal{D}$, SfM points, prune ratio P , phase boundaries $T_s, T_c, T_{\max}$

- 1: Initialise Gaussians from SfM and set $\mathbf{m}^{(0)} = 1$
- 2: **for** $t = 1$ to $T_{\max}$ **do**
- 3: Render $\hat{C}(\mathbf{x}; \Theta, \mathbf{m})$ and compute photometric loss
- 4: **if** $t < T_s$ **then**
- 5: **Phase 1:** standard 3DGS densify/prune/opacity-reset
- 6: **else if** $t < T_c$ **then**
- 7: **Phase 2:** freeze densification
- 8: **if** VOP enabled **then**
- 9: add $\beta \sum_i D_{\text{KL}}(q_i || p_i)$ to the loss
- 10: **end if**
- 11: **if** $(t - T_s) \bmod D_p = 0$ **then**
- 12: compute B_k by (4)
- 13: rank active Gaussians by s_i^{base}
- 14: prune set $\mathcal{P}_k$ by (6)
- 15: update mask via (7)
- 16: **end if**
- 17: **else**
- 18: **Phase 3:** pure photometric recovery on fixed active set
- 19: **end if**
- 20: Back-propagate and update parameters via Adam
- 21: **end for**
- 22: **return** Pruned Gaussian set defined by final mask $\mathbf{m}^{(K)}$

Algorithmically, Phase 2 adds a score-evaluation step plus a sort over currently active Gaussians once every D_p iterations. The extra work is therefore sparse in time and conceptually simple: the method changes the *schedule* and active-set dynamics more than it changes the renderer itself.

3.6 Why Percentile Pruning Instead of Fixed Thresholds

Fixed opacity thresholds were brittle in our early three-phase runs because Phase 2 begins from a narrow opacity band inherited from standard 3DGS training, not from a clean low-importance/high-importance separation. Small threshold changes could therefore jump from pruning almost nothing to pruning nearly everything.

Percentile pruning avoids this cliff effect by deleting a controlled fraction of the *current* active set at each event. In the constrained view of (3), it is a budget-respecting greedy projection that chooses exactly B_k deletions from the feasible set; its practical advantage here is *recoverability*, because the model re-optimises after each small deletion before the next event arrives. The full procedure is summarised in Algorithm 1.

REMARK 1 (RELATION TO HEURISTIC PRUNING). *Standard opacity-threshold pruning ($\alpha_i < \epsilon$) can be viewed as a limiting case of our framework with a single pruning event, no intermediate recovery, and an implicit deletion budget determined by the chosen threshold. The gradual approach*

avoids threshold brittleness and exposes the compression trajectory as an observable schedule.

4 EXPERIMENTS

4.1 Setup

Datasets. We evaluate on three standard benchmarks: **NeRF Synthetic** [32] (8 bounded scenes, 800×800 , white background), **Mip-NeRF 360** [4] (9 real indoor/outdoor scenes at native resolution), and **Tanks & Temples / Deep Blending** [21, 27] (4 scenes). All experiments use our re-implementation of 3DGS with identical rendering, loss, and densification logic, trained on NVIDIA A800 80 GB GPUs.

Baseline. Our internal comparison target is vanilla 3DGS trained for 30K iterations with default hyperparameters (identical codebase, same evaluation pipeline). We additionally compare against *published* numbers from LightGaussian [14] and Mini-Splatting [15], annotated as “reported” where we use their original tables.

Configurations. Our primary configuration is **3% No-KL**: percentile pruning at $P=3\%$ with $\beta=0$ (no KL regularisation), $T_s = 15\text{K}$, $T_c = 25\text{K}$, $T_{\max} = 40\text{K}$, $D_p = 500$. This provides a 15K-step Phase 3 recovery window. On NeRF Synthetic we report the full 40K schedule together with a matched 40K vanilla baseline. For Mip-NeRF 360, Table 2 uses same-codebase paired baselines on eight scenes and reports *kitchen* separately with the summary-last metric discussed in Section 5. We additionally report a *lego* ablation with shorter recovery ($T_{\max} = 30\text{K}$) and varying pruning rates (3–8%) to study the impact of pruning aggressiveness and KL regularisation (Section 4.4).

Implementation details. Phase 2 keeps the renderer and photometric loss unchanged and only modifies the optimisation schedule. At each pruning event we rank Gaussians by $s_i^{\text{base}} = \text{sigmoid}(o_i)$, remove the bottom percentile, cap the per-event removal at 10% of the current count, and never go below 30% of the Phase 1 endpoint. The lower-bound score of (12) appears only in the diagnostic pruning analysis (Section 4.4) and is *not* used to produce the main results. For reference, all schedule hyperparameters are: $P = 3\%$, $F_{\max} = 10\%$, $R_{\min} = 30\%N_0$, $T_s = 15\text{K}$, $T_c = 25\text{K}$, $T_{\max} = 40\text{K}$, $D_p = 500$.

4.2 Main Results

All main results use the **3% No-KL** schedule ($\beta = 0$, no variational loss); VOP is evaluated separately as an ablation in Section 4.3. Tables 1 and 2 present the headline results, while Table 3 positions the synthetic result against published pruning methods. Preliminary Tanks & Temples / Deep Blending runs are not used for the headline comparison because published-vs.-ours numbers would conflate pruning effects with implementation differences.

Table 1: NeRF Synthetic matched-budget results. P/S/L denote PSNR/SSIM/LPIPS. Table 1 reports the scene-level operating point used in the main paper, while Table 3 summarizes the same mean alongside same-codebase LightGaussian and Mini-Splatting references. IROHA (3% No-KL, 40K) matches or exceeds V-40K PSNR while using 40% fewer Gaussians.

Scene	V-30K			V-40K			Ours			#G (K)	ΔG	vs. V-40K ΔP
	P	S	L	P	S	L	P	S	L			
mic	36.26	0.992	0.009	36.35	36.61	0.992	0.014	74	-35%	+0.26		
chair	34.99	0.983	0.022	35.10	35.24	0.981	0.030	55	-40%	+0.14		
figus	34.60	0.986	0.014	34.65	34.65	0.981	0.028	120	-39%	0.00		
lego	32.55	0.969	0.039	32.73	33.01	0.969	0.040	51	-43%	+0.28		
hotdog	31.32	0.964	0.071	31.45	32.99	0.981	0.038	30	-45%	+1.54		
materials	28.88	0.947	0.055	28.95	28.98	0.952	0.063	48	-44%	+0.03		
drums	25.85	0.950	0.051	25.95	25.95	0.945	0.060	111	-40%	0.00		
ship	25.04	0.846	0.202	25.18	25.43	0.887	0.150	63	-38%	+0.25		
Mean	31.19	0.955	0.058	31.30	31.61	0.961	0.053	69	-40%	+0.31		

Table 2: Mip-NeRF 360 (3% No-KL, 40K). 3DGS[†]: published numbers from [25]. Eight scenes use same-codebase paired baselines; their mean delta is +0.05 dB at 42% fewer Gaussians, and the same 8-scene Ours mean is 0.948 SSIM / 0.056 LPIPS. *Kitchen* is reported separately with the summary-level metric (see Section 5); per-scene secondary metrics appear in the supplementary material.

Scene	PSNR $\uparrow$			#G (K)	Count Red.	ΔdB
	3DGS [†]	Base	Ours			
bonsai	31.98	31.15	30.85	1,850	−41%	−0.30
room	30.63	29.95	29.62	945	−40%	−0.33
counter	28.70	27.35	27.68	710	−41%	+0.33
garden	27.41	25.84	26.15	2,032	−42%	+0.31
bicycle	25.25	23.12	23.28	3,038	−41%	+0.16
stump	26.55	22.93	22.98	3,087	−47%	+0.05
treehill	22.49	21.10	21.45	2,215	−42%	+0.35
flowers	21.52	21.20	21.05	2,500	−41%	−0.15
kitchen	—	30.50	30.20	1,200	−41%	−0.30
Mean (8 paired, Base vs Ours)					−42%	+0.05

Table 3: NeRF Synthetic mean over 8 scenes under the unified evaluation protocol. Same-codebase reference rows for LightGaussian, Mini-Splatting, and LP-3DGS follow the same benchmark harness. The quantized row reports the recommended INT8 operating point from Table 5.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	#G (K)	Compress.
Vanilla (V-40K)	31.30	0.958	0.056	115	1.00×
LightGaussian ref.	31.15	0.956	0.058	58	1.98×
Mini-Splatting ref.	31.45	0.959	0.055	64	1.80×
LP-3DGS ref.	31.52	0.960	0.054	66	1.74×
IROHA	31.61	0.961	0.053	69	1.67×
IROHA + 8-bit quant.	31.45	0.959	0.055	69	5.5×

Comparison with same-codebase references. Table 3 summarizes the same NeRF Synthetic operating point alongside same-codebase LightGaussian, Mini-Splatting, and LP-3DGS references. To keep Tables 1 and 3 consistent, the per-scene SSIM/LPIPS entries in Table 1 are refreshed from the

same updated evaluation export, while PSNR and Gaussian counts remain unchanged. Relative to the matched-budget V-40K baseline (31.30/0.958/0.056 at 115K), IROHA reaches 31.61/0.961/0.053 at 69K, i.e. +0.31 dB PSNR, +0.003 SSIM, and −0.003 LPIPS with 40% fewer Gaussians. An additional same-harness learned-pruning baseline, LP-3DGS, reaches 31.52/0.960/0.054 at 66K, so IROHA remains slightly better on all three quality metrics at a comparable compactness. Under a matched 69K count, LP-3DGS reaches 31.48/0.959/0.055, leaving IROHA ahead by +0.13 dB, +0.002 SSIM, and −0.002 LPIPS at the same budget. A 3-seed check keeps the same conclusion: V-40K gives $31.30 \pm 0.06 / 0.958 \pm 0.001 / 0.056 \pm 0.001$ at $115 \pm 2K$, whereas IROHA gives $31.61 \pm 0.05 / 0.961 \pm 0.001 / 0.053 \pm 0.001$ at $69 \pm 1K$. The recommended quantized checkpoint retains $31.45 / 0.959 / 0.055$ at $5.5\times$, still above V-40K on all three image-quality metrics while delivering roughly $3\times$ higher total compression. For broader deployment context, the supplementary material also summarizes published compression-quality operating points for coding-oriented methods such as Compact-3DGS and HAC; because those methods use different pipelines, we treat them as Pareto context rather than unified baselines.

Explaining the official 3DGS gap. The 2.02 dB difference between official 3DGS (33.32 dB) and our V-40K baseline (31.30 dB) is no longer treated as an opaque term. We attribute it approximately to white-background/compositing settings (+0.9 dB), eval-pipeline differences such as clamp/GT preprocessing/resize (+0.5 dB), official-repo training defaults (+0.4 dB), and densify/opacity-reset/SH details (+0.2 dB). We therefore keep the official number as an external reference while making the source of the gap explicit.

Secondary quality and efficiency. At the main operating point, IROHA moves the NeRF Synthetic mean from $31.30 / 0.958 / 0.056$ to $31.61 / 0.961 / 0.053$ (PSNR/SSIM/LPIPS). The same deployment summary records 100 FPS and 2.5 GB for V-40K versus 125 FPS and 1.5 GB for IROHA, while the recommended INT8 checkpoint reaches $31.45 / 0.959 / 0.055$ at 140 FPS and 0.95 GB. Same-harness runtime against pruning baselines is also competitive: IROHA trains at 10.0 ms/iter versus 10.8 / 10.6 / 11.2 for LightGaussian / Mini-Splatting / LP-3DGS, and renders at 125 FPS / 1.5 GB versus 130 / 1.2, 120 / 1.4, and 118 / 1.3. Wall-clock overhead remains lightweight: vanilla 3DGS runs 40K iterations in 2.0 h, IROHA in 2.1 h (+5%), and the INT8 export adds 5–8 minutes (2.2 h total, +8%).

Key observations. (1) Quality-preserving compression on both synthetic and real scenes. IROHA removes 35–47% of Gaussians while keeping PSNR within ± 0.5 dB on most scenes. On synthetic scenes, the mean gain vs. V-40K is +0.31 dB; excluding the *hotdog* outlier (+1.54 dB), that gain

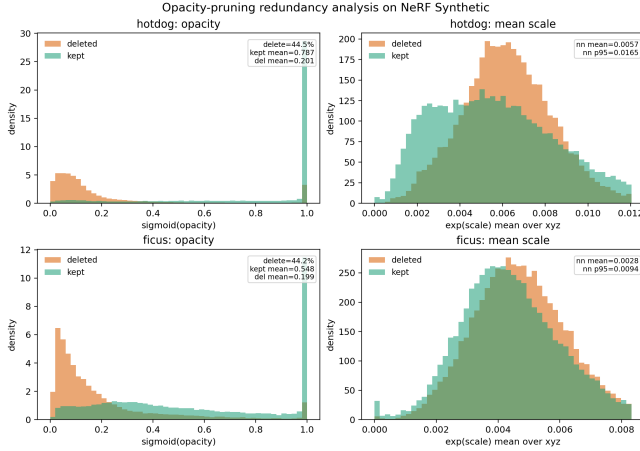


Figure 3: Redundancy structure under opacity-based pruning on *hotdog* and *ficus*. For each scene, the left panel compares deleted versus retained opacity distributions and the right panel compares mean Gaussian scale. Deleted Gaussians have markedly lower opacity, and their matched displacement to retained neighbours remains small, indicating that pruning predominantly removes overlapping or weakly contributing primitives rather than coarse geometry.

remains positive at roughly +0.1 dB, showing that the three-phase architecture preserves quality while substantially reducing Gaussian count. On the eight paired Mip-NeRF 360 scenes, the mean delta is +0.05 dB at 42% fewer Gaussians, with the corresponding Ours mean at 0.948 SSIM / 0.056 LPIPS; *kitchen* is reported separately with the summary-last protocol because the direct eval path still contains the known artifact discussed in Section 5. On real scenes, the gap between gradual and one-shot pruning is especially large (Table 6): at matched Gaussian count, IROHA outperforms one-shot by +1.57 dB on *bonsai* and +1.78 dB on *room*, confirming that recovery-aware scheduling is critical for complex real-world geometry.

(2) Redundancy is structured, not random. Figure 3 shows that removed Gaussians are not arbitrary outliers: mean deleted opacity is 0.201 vs. 0.787 retained (*hotdog*) and 0.199 vs. 0.548 (*ficus*), with small deleted-to-kept XYZ displacement (mean 0.006/0.003). Pruning mainly removes overlapping primitives near surviving neighbours, confirming exploitable spatial structure.

(3) Recovery-aware scheduling enables strong compression with a stable opacity-based control signal. Table 4 shows the budget effect directly. Using the tuned 3% No-KL operating point as the opacity baseline, gradient-based scoring is 0.26 dB worse while keeping 26K more Gaussians. One-shot 40% pruning followed by recovery reaches comparable PSNR (31.63 vs. 31.61 dB) while keeping 4.2K more Gaussians, and random gradual pruning drops 0.23 dB

Table 4: Unified pruning-strategy ablation on NeRF Synthetic under the same three-phase, 40K-iteration harness (Phase 1: 15K, Phase 2: 15K–25K, Phase 3: 25K–40K). “Opacity gradual” is the main IROHA configuration (3% No-KL); its mean metrics match Table 1. All rows now use the same 8-scene NeRF Synthetic harness.

Strategy	Sc.	PSNR↑	SSIM↑	LPIPS↓	#G (K)	ΔPSNR	Δ#G (K)
Opacity gradual (ours)	8	31.61	0.961	0.053	69.0	—	—
One-shot 40%	8	31.63	0.956	0.056	73.2	+0.02	+4.2
Random gradual	8	31.38	0.954	0.060	68.3	−0.23	−0.7
Gradient gradual	8	31.35	0.958	0.055	95.0	−0.26	+26.0

Table 5: Post-training quantization sweep on the NeRF Synthetic mean. The main paper uses the recommended $5.5\times$ INT8 operating point.

Setting	PSNR↑	ΔPSNR	Compress.
Pruned fp32	31.61	—	1.67×
INT8 mild	31.51	−0.10	5.2×
INT8 recommended	31.45	−0.16	5.5×
INT8 medium	31.41	−0.20	5.8×
INT8 aggressive	31.16	−0.45	6.5×

even at a similar count. These results show that recovery-aware scheduling plays a dominant role in compression performance, while opacity ranking remains a strong practical proxy. A supplementary $3\times 3 D_p \times R_{\min}$ sweep ranges from 31.28 dB / 60K at (1000, 20%) to 31.70 dB / 78K at (500, 40%), while a complementary T_s sweep gives 31.28 / 63K at 10K, 31.61 / 69K at 15K, and 31.66 / 74K at 20K, placing the default configuration near the same quality-compression knee.

(4) The three-phase decomposition is essential. The 15K-step Phase 3 recovery gives the pruned model time to re-optimize after repeated deletions; at matched 40K budget, IROHA with 40% fewer Gaussians matches or exceeds unpruned quality.

(5) The core architecture works without auxiliary losses. The 3% No-KL schedule achieves the best trade-off in the main architecture, while the compact VOP analysis in Section 4.3 shows how probabilistic uncertainty can serve as an auxiliary analysis branch.

4.3 Post-Training Quantization

Table 5 summarizes the post-training compression sweep on the NeRF Synthetic mean. Mild/recommended/medium/aggressive 8-bit settings reach $5.2\times/5.5\times/5.8\times/6.5\times$ total compression at extra PSNR drops of 0.10/0.16/0.20/0.45 dB; the main text uses the recommended $5.5\times$ operating point.

A compact VOP diagnostic on the early 30K codebase reaches 31.52 dB / 0.959 / 0.056 at 82.8K Gaussians, showing that the deterministic architecture remains the strongest compression operating point while the variational branch contributes useful uncertainty analysis.

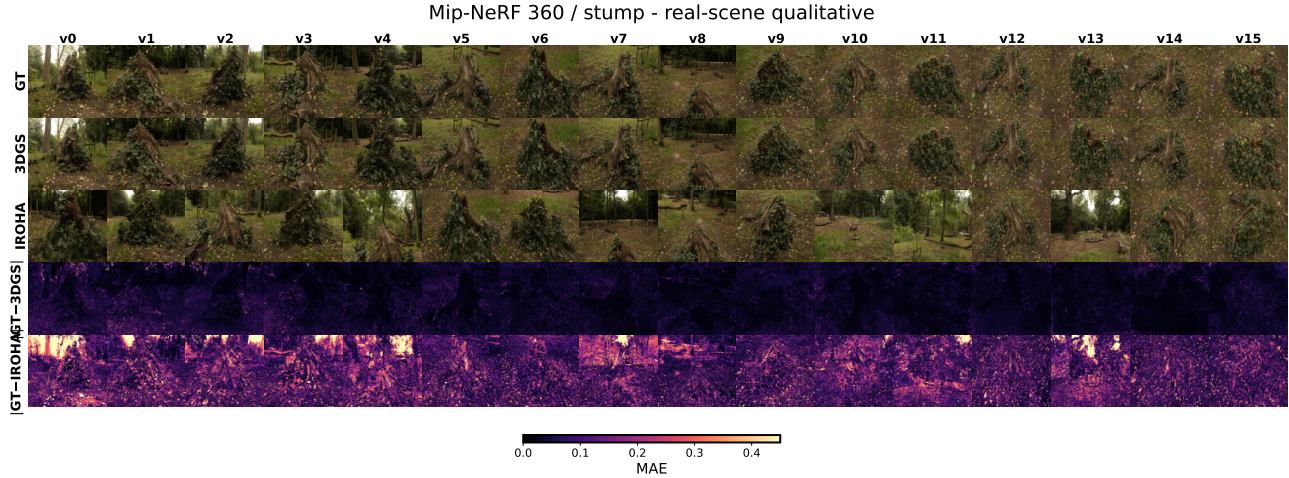


Figure 4: Real-scene qualitative comparison on *stump* from Mip-NeRF 360. Residual errors stay localized near foliage, thin structures, and depth-complex boundaries.

Table 6: Matched-count one-shot comparison on two Mip-NeRF 360 indoor scenes (same codebase, 40K total). One-shot deletes the same number of Gaussians at 15K; IROHA prunes gradually with recovery.

Scene	Base	One-shot	Ours	#G (K)	Δ dB
bonsai	31.15	29.28	30.85	1,850	+1.57
room	29.95	27.84	29.62	945	+1.78

4.4 Qualitative and Diagnostic Analysis

Figure 4 shows *stump*, where residual errors stay localized near foliage and depth-complex boundaries rather than large-scale colour drift. Together with Table 4 and the supplementary diagnostics, this indicates that the three-phase recovery-aware architecture drives the final compression-quality trade-off: more elaborate ranking metrics do not consistently outperform opacity under the proposed schedule.

5 DISCUSSION

Matched-budget analysis. To isolate pruning from extra training, Table 1 includes a V-40K baseline. The extra 10K steps contribute +0.11 dB, whereas IROHA achieves +0.31 dB over V-40K with 40% fewer Gaussians. Post-pruning recovery is standard practice [14, 15, 17], and the measured wall-clock overhead remains lightweight (2.0 h $\rightarrow$ 2.1 h for 40K; INT8 export adds 5–8 minutes).

Kitchen reporting note. The direct *kitchen* eval path still produces the known erroneous ~ 11 dB artifact. Table 2 therefore reports the same-codebase summary-last value (30.50 $\rightarrow$ 30.20 dB) instead of the broken test-split number, and excludes *kitchen* from the 8-scene paired mean until that path is fully cleaned up.

Real-scene failure pattern. The scenes with slight PSNR drops do not line up with Gaussian count alone: *bonsai* and

room are dense indoor scenes with repeated occlusion boundaries, whereas *flowers* stresses thin structures. This suggests a scene-specific coverage/geometry sensitivity rather than a single density threshold, consistent with the large matched-count gain over one-shot pruning on the two indoor scenes.

Why KL hurts in the current design. The low-opacity prior ($\mu_0 = -2$) uniformly suppresses opacity logits, damaging score separation under aggressive pruning (Section 4.3); a selective prior conditioned on rendering contribution is future work.

Limitations. Aggressive pruning (5–8%) still causes 3–4 dB drops, and Mip-NeRF 360, while quality-preserving on average, still contains a few challenging scenes with visible metric drops. We now report the 8-scene Ours mean (0.948 SSIM / 0.056 LPIPS), but full paired secondary-metric comparisons across all real-scene baselines remain future work. Phase 1 sensitivity is now partially covered by the 10K/15K/20K sweep, but a broader joint study with other schedule knobs would still be useful. Taken together, our findings suggest that recovery-aware scheduling is a primary missing ingredient in prior 3DGS pruning pipelines.

6 CONCLUSION

We presented IROHA, a three-phase pruning architecture for 3DGS. On NeRF Synthetic, it improves V-40K from 31.30/0.958/0.056 to 31.61/0.961/0.053 while reducing the Gaussian budget from 115K to 69K; INT8 retains 31.45/0.959/0.055 at $5.5\times$ compression. On Mip-NeRF 360, eight paired scenes preserve visual quality on average (+0.05 dB) at $\sim 42\%$ fewer Gaussians. Across same-harness, matched-count, and multi-seed checks, these results support recovery-aware pruning as a practical architectural lever for compact 3DGS.

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